Conference Paper Title*

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***Abstract*—Provide a brief summary of: What problem do you aim to address? Why is this problem an important application? What is your proposed approach? How does the proposed approach compare to prior work? Describe what dataset used to train and evaluate the proposed machine learning models used and provide sources. Describe the design of the front- and back-end of the application. How will you evaluate the proposed machine learning models including metrics? What are the expected findings? What impacts does this project have on the machine learning field broadly?**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Motivation: The problem of flight delays in the U.S. continues to exist. Flight delays impact many passengers every year and negatively impact them in many different ways, including causing passengers to miss connecting flights, causing higher costs, making it harder to achieve reliable travel, etc. Currently, there are very few tools that provide predictive information on flight delays; most only provide a historical record of what the airline has done previously or update the passenger with real-time information. The other issue is that passengers generally are not familiar with why certain flights are delayed and rarely know what to do to obtain a better travel option. In light of this issue, we propose an application using machine learning to allow the passenger to enter their departing airport, arriving airport, and date of travel and receive a prediction of both the probability of flight delay and the predicted duration of the flight delay based on the individual airline being used for the flight. In addition, users would also receive an airline recommendation based on the risk of flying with that airline so that they can make better travel choices.

Technical focus: We define this problem as a classification task (predicting whether a flight will be delayed) and a regression task (predicting the duration of the delay). To capture the complex relationships among numerous factors such as weather, airline operations, and airport congestion, we employed a combination of baseline models and ensemble models, including logistic regression and forest models. One of the key contributions of our approach lies in the use of five structured delay categories from the BTS dataset - airline, weather, national aviation system, security check, and delayed aircraft delay - as core features. This not only enables us to predict the delay situation but also explains the underlying causes and provides users with more meaningful suggestions.

Prior Work: Delay prediction through previous research has mainly used machine learning or statistical analysis to analyse numerous delays based upon past events to develop delay patterns. This research attempts to find the most significant factors contributing to delays as well as improve delay prediction accuracy. However, almost all other delay prediction research used single outcome variables for delay prediction purposes and did not define a method to classify the cause of each delay) and did not provide an avenue for extending research efforts into user-friendly applications. Therefore, our research will be different in that we define multiple delay type classifications, and bring together delay prediction with a user-friendly decision-support system to aid in decision making.

Machine Learning Pipeline: Our pipeline includes data collection, exploratory data analysis (EDA), preprocessing, model training, model evaluation, and deployment. We worked with 2025 Bureau of Transportation Statistics (BTS) flight data to get information on the airline and route, and delay features to include in our models. The delay features we used included features from carrier delays, weather delays, NAS delays, security delays, and late aircraft delays. We also used EDA to examine the delay distributions for routes, airlines, and regions to find patterns related to the delays for each type. After identifying the missing values in the data and encoding the categorical variables, we built classification and regression algorithms to predict the likelihood of or length of a delay, with the training/testing split giving us an accurate assessment

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of the models. The performance of the model was evaluated with accuracy, precision, recall, F1 score, and RMSE. Once the best-performing model was identified, it was deployed through a Streamlit application that allows the user to input the desired route and date, and to subsequently receive the likelihood of the delay occurring, which is based on the route and date the user provided, along with any other likely contributing factors to the delay.

Impact: The objective of this study is to assist users in making flight travel decisions by providing them with predictive analysis, enabling them to gain a deeper understanding of flight delays. This will help users better identify the causes of delays, allowing them to choose more reliable travel options and optimize their travel plans. Although this study will provide users with insights into the causes of delays in specific regions, due to the large number of large airports, data bias may affect the accuracy of the model, and thus the results may not be applicable to the entire range of airport codes. Therefore, we will provide users with probability prediction estimates and simultaneously show them the uncertainty and its influencing factors.

II. Background

Review prior work relevant to your project and provide a brief background, discuss how your proposed work is similar and different from prior work, and cite sources in bibtex. Google Scholar provides bibtex code. Place the bibtex code in the references.bib file in Overleaf and cite the source (e.g., [**?**]). Based on prior work and results, how do you anticipate your proposed approach will compare to prior approaches?

A. Back-End

The Back-End of the web application performs data exploration and preprocessing, model training and evaluation, and launches an application that deploys that trained models. Think of the back-end as the high-level functionality behind website user interface elements (e.g., menus, buttons). For instance, the backend of HW2 trained models when a user pressed a button.

1) Data Collection, Exploration & Processing: We use the 2025 BTS Airline On-Time Performance dataset obtained from the official U.S. Bureau of Transportation Statistics (BTS) TranStats Airline On-Time Performance Reporting download portal. This is an official U.S. government data source that provides public flight performance records, making it an appropriate and reliable dataset for our project. After merging the 12 monthly files for 2025, the dataset contains 6,390,044 rows and 23 columns, which provides broad coverage across airlines, origins, destinations, and dates. It is a structured tabular dataset with mixed data types. It includes temporal features (YEAR, QUARTER, MONTH, FL_DATE), categorical features (OP_UNIQUE_CARRIER, ORIGIN, ORIGIN_CITY_NAME, ORIGIN_STATE_NM, DEST, DEST_CITY_NAME, DEST_STATE_NM), numerical features (DEP_DELAY, DEP_DELAY_NEW, AIR_TIME, DISTANCE, CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY, etc.), and a binary target-related variable (DEP_DEL15). In total, the merged dataset has 11 float columns, 4 integer columns, and 8 object columns. We will use this dataset to build a model to investigate whether it is possible to predict flight delays in the United States based on weather, airline, departure or arrival cities, and operational factors. We will also use this to create a system that helps users choose more reliable airlines when traveling. The dataset already includes ground truth labels. We will use DEP_DEL15 as the classification label and DEP_DELAY_NEW as the regression label.

We conducted an analysis of the types and causes of delays across various airlines. First, we analyzed the cleaned dataset, which contains only flights with complete data, and calculated the average forward departure delay time (DEP_DELAY_NEW) and the percentage of flights delayed by 15 minutes or more (DEP_DEL15) for each airline. As shown in Figure 1, our calculations reveal significant differences among airlines. The average forward departure delay ranges from approximately 44.82 to 85.48 minutes, with OH, F9, G4, AA, and OO exhibiting higher values, whereas AS, HA, MQ, and WN show relatively lower values. Meanwhile, the proportion of flights delayed by 15 minutes or more ranges from approximately 69.92 percentage to 85.97 percentage, with WN, G4, OH, NK, and AA having higher rates, while AS, YX, UA, and HA have relatively lower rates. Based on these statistics, we have created a bar chart showing the average delay time per flight by airline (Figure 2) and a bar chart showing the proportion of delayed flights by airline (Figure 3).

We also generated correlation heatmaps for five different causes of delays. Based on the heatmaps (Figure 4), we found that most types of delays are positively correlated with one another. In particular, delays caused by airline operational issues are closely correlated with weather-related delays, and airline system delays are also closely correlated with delays caused by late-arriving aircraft. In contrast, security check delays show weaker correlations with most other types of delays. This suggests that several operational delay causes often co-occur and should be considered together in later analysis.

During the preprocessing stage, we first merged the 12 monthly CSV files into a single dataset and examined the overall structure, column types, missing values, and descriptive statistics. We then removed rows with missing values from the key delay-related columns and performed data cleaning. Specifically, if any of the variables DEP_DELAY, DEP_DELAY_NEW, DEP_DEL15, CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, or LATE_AIRCRAFT_DELAY contained NAN values, we excluded the entire column of data. This is because the primary research question of our project is: Can we use weather, airline, departure/arrival cities, and operational factors to predict whether U.S. flights will be delayed? Therefore, we only need to retain data related to these

OP_UNIQUE_CARRIER	DEP_DEL15	CARRIER_DELAY	WEATHER_DELAY	NAS_DELAY	SECURITY_DELAY	LATE_AIRCRAFT_DELAY	DEP_DELAY_NEW	Total_Flights	Delay_Ratio (%)	Avg_Delay_Per_Flight (min)	Total_Delay (min)	Delay_Contribution (%)
AA	0.8039086033355380	5950136.0	1018707.0	2802063.0	22784.0	8204180.0	78.50232808022920	217776	80.39086033355380	78.50232808022920	17997870.0	18.188096206244800
AS	0.6991692796351350	706849.0	90450.0	725963.0	11378.0	995203.0	44.82306470660100	49114	69.91692796351350	44.82306470660100	2529843.0	2.5565818550025600
B6	0.7938098501546610	1115014.0	107499.0	1146168.0	5963.0	1720214.0	71.23493674637430	53989	79.38098501546610	71.23493674637430	4094858.0	4.138138082723730
DL	0.7839724402778560	5532880.0	638915.0	2745774.0	8011.0	3948482.0	68.46058056135990	177070	78.39724402778560	68.46058056135990	12874062.0	13.010132766886300
F9	0.8011569802401050	1128160.0	117405.0	776898.0	0.0	2035903.0	79.84314416637290	48229	80.11569802401050	79.84314416637290	4058366.0	4.101260385154060
G4	0.8198619224641530	724784.0	280740.0	444995.0	9505.0	961703.0	79.55751460435480	28245	81.98619224641530	79.55751460435480	2421727.0	2.4473231366412000
HA	0.723814862267777	407443.0	22954.0	8262.0	2320.0	223137.0	48.603699551569500	12488	72.3814862267777	48.603699551569500	664116.0	0.6711352899041080
MQ	0.7343620527740440	697490.0	413966.0	837370.0	3961.0	1497538.0	54.41985379417020	54307	73.43620527740440	54.41985379417020	3450325.0	3.4867927728565400
NK	0.8117026798153360	684117.0	77422.0	1268373.0	7809.0	685360.0	71.28117573141750	36607	81.17026798153360	71.28117573141750	2723081.0	2.751862259556110
OH	0.8117048784692300	1515378.0	363585.0	696172.0	7803.0	2641543.0	85.47760730908470	58010	81.17048784692290	85.47760730908470	5224481.0	5.279700489874510
OO	0.7660333396600590	6028462.0	1862736.0	2271880.0	14630.0	2545757.0	77.31434443128200	152791	76.60333396600590	77.31434443128200	12723465.0	12.857944050978700
UA	0.7185142155536050	2808929.0	540245.0	3585270.0	601.0	3972691.0	65.1249661691608	151489	71.85142155536050	65.1249661691608	10907736.0	11.023023933405400
WN	0.8597261788667830	3874336.0	426305.0	2879169.0	34736.0	7504670.0	56.38308851195180	260462	85.97261788667830	56.38308851195180	14719216.0	14.874788888268300
YX	0.7126948775055680	1056808.0	254734.0	1588395.0	3020.0	1662014.0	59.91660480079190	64656	71.26948775055680	59.91660480079190	4564971.0	4.613219882503730

Fig. 1. Airline-Level Delay Metrics and Delay Cause Statistics in the Cleaned Delayed-Flight Subset.

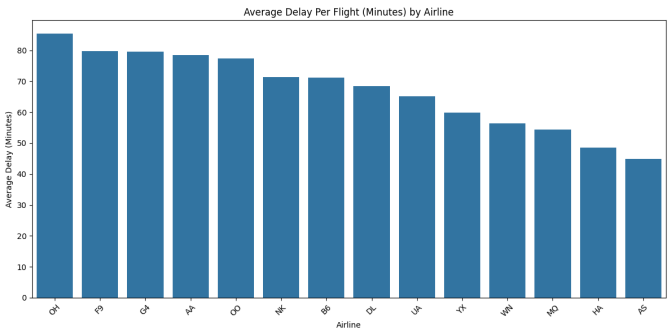

Fig. 2. Average positive departure delay per delayed flight by airline in the cleaned delayed-flight subset.

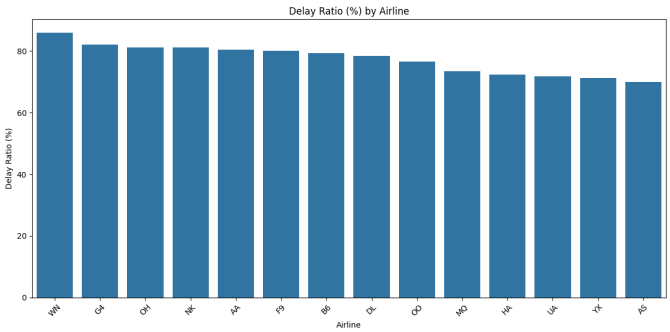

Fig. 3. Proportion of flights with departure delays of 15 minutes or more by airline in the cleaned delayed-flight subset.

factors and do not need to retain information on flights that departed on time or ahead of schedule. Since our delay data contains negative values (early departures), we truncated these negative values to 0 to ensure that the delay measurements are valid non-negative values. We also created variables for each airline, including delay rate, average delay time per flight, total delay time, and the percentage of delayed flights, to understand the relationship between airlines and delays. This provides valuable insights for the subsequent research in our project. Moreover, we also categorized the continuous variable "DISTANCE" into short-haul and long-haul flights. Flights with a distance of 500 miles or less were classified as short-haul, while those with a distance greater than 1,500 miles were classified as long-haul.

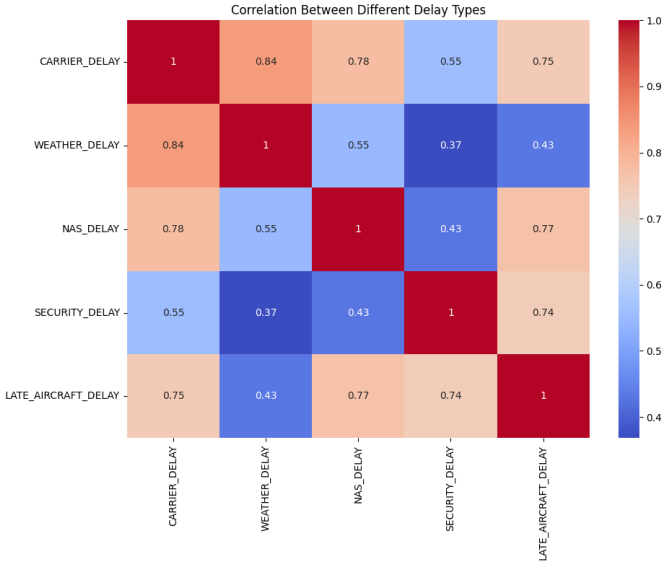

Fig. 4. Correlation heatmap of the five BTS delay-cause variables in the cleaned delayed-flight subset.

2) Methods and Model Training: What machine learning techniques are you planning to apply or improve upon and how? Provide justification for why the machine learning approach is well-suited to address the problem. What are the model inputs and outputs? Describe the training and validation procedure including the train/test split. State model parameter settings and justification. Discuss the methods used to avoid model overfitting and underfitting.

Make sure to describe them in detail and provide enough context for the reader to understand the methods at least at a high level. Provide any background that is necessary for that.

3) Model Evaluation: escribe the planned experiments, expected outcomes, and error analysis. State the evaluation methods for comparison (at least three required), metrics (at least two required), and hyperparameter settings. The evaluation methods consist of all methods used for comparison. The evaluation metrics assess how well the models perform and can be used to identify over- and underfitting. State whether the metrics used are commonly used for evaluating similar tasks and cite sources.

Make sure that the setup is described in enough detail

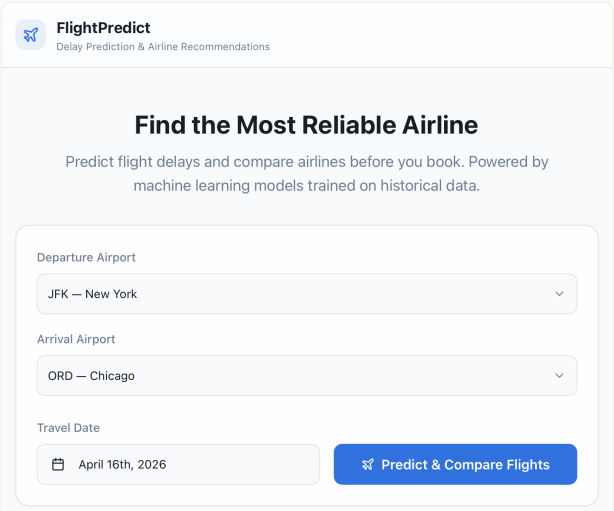

Fig. 5. Home page input interface of the Streamlit application.

for someone else to reproduce your results. Also, if you have an experimental project, make sure to provide a detailed experimental analysis. Things you should consider including: train/test performance, learning curves, model samples, error analyses, ablation analyses, etc.

Context: Explain how you build upon previous work and how your expected results compare to what has been done previously (be realistic in your prediction).

4) Model Deployment: Describe the application of proposed project including user inputs and model outputs. Discuss the importance of the application as well as ethical or societal implications of its usage with everyday users.

B. Front-End (Streamlit)

The front-end of the proposed web application is developed using the Streamlit framework, which enables rapid integration of machine learning models with an interactive and user-friendly interface. The goal of the application is to translate predictive modeling results into actionable insights that assist users in making better travel decisions.

1) Target Population: The primary target users of this application are travelers within the United States, including business travelers, students, and tourists, who seek to reduce the risk of flight delays. These users typically do not have a technical background in machine learning; therefore, the application is designed to be intuitive, requiring minimal input while providing clear and interpretable outputs.

2) User Interface Layout: The web application is structured into two main sections: an input interface and a results dashboard.

The input interface (home page) allows users to specify flight details, including departure airport, arrival airport, and travel date. The design follows a clean and minimal layout to reduce cognitive load and ensure ease of use.

Once the user submits the input, the application transitions to a results dashboard that presents model predictions and

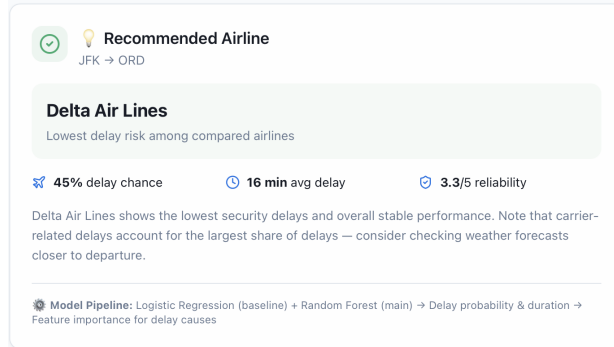

Fig. 6. Results dashboard displaying prediction outputs and airline recommendations.

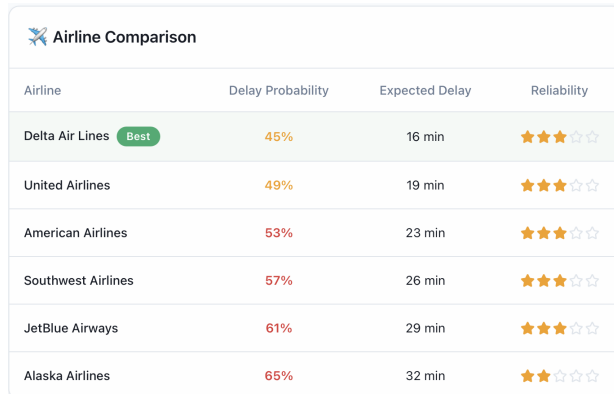

Fig. 7. Airline comparison table with delay probability and reliability ratings.

insights in a structured format. The dashboard includes the following components:

- **Recommended Airline Panel:** Highlights the airline with the lowest predicted delay risk, along with delay probability, expected delay duration, and a reliability score.
- **Airline Comparison Table:** Displays multiple airlines with their corresponding delay probability, expected delay time, and reliability ratings, enabling direct comparison.
- **Delay Comparison Visualization:** A bar chart comparing delay probability and expected delay duration across airlines.
- **Delay Causes Breakdown:** A donut chart illustrating the contribution of different delay factors, including carrier, weather, NAS, security, and late aircraft delays.

This layout ensures that users can quickly interpret results and make informed decisions.

3) User Inputs and Outputs: The application utilizes simple and structured input components:

- Dropdown menus for selecting departure and arrival airports
- A date picker for selecting the travel date
- A button labeled *"Predict & Compare Flights"* to trigger model execution

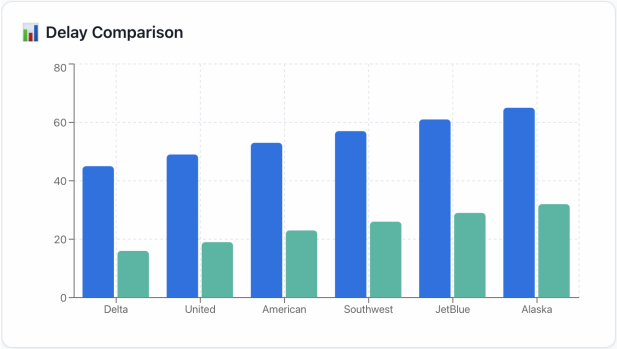

Fig. 8. Bar chart comparing delay probability and expected delay duration across airlines.

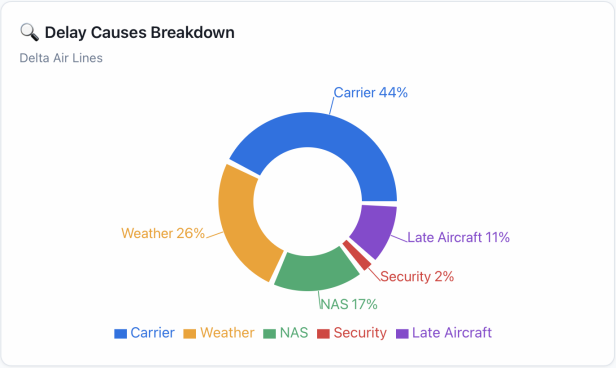

Fig. 9. Donut chart illustrating the breakdown of delay causes by contributing factor.

After the user submits the inputs, the application generates the following outputs:

- Predicted probability of flight delay (classification output)
- Predicted delay duration in minutes (regression output)
- A ranked comparison of airlines based on delay risk
- A recommended airline based on combined prediction metrics
- An explanation of delay causes using feature importance

These outputs are presented visually and textually to enhance interpretability and usability.

4) Front-End and Back-End Integration: The Streamlit application integrates the front-end interface with the machine learning pipeline in a seamless manner. When the user clicks the prediction button, the selected inputs are passed to the backend model pipeline.

The backend consists of both a baseline Logistic Regression model and a primary Random Forest model. The Random Forest model is used to generate the final predictions, including delay probability and expected delay duration. Additionally, feature importance scores derived from the model are used to explain the contributing factors behind flight delays.

Once the model inference is complete, Streamlit dynamically updates the interface to display the results in the form of tables, charts, and recommendation panels, as shown in Figures 8 and 9. A loading indicator can be incorporated to inform users that the model is processing their request, improving the overall user experience.

5) Application Design Considerations: The application is designed to be clear, concise, and actionable. It minimizes user input requirements while maximizing the value of the output. By combining prediction, comparison, and explanation in a single interface, the system transforms raw historical data into meaningful decision support.

Overall, the application demonstrates how machine learning models can be effectively deployed in a real-world scenario to improve user decision-making and enhance travel planning.

III. Risk & Mitigation

The disparity in categories of flight delays presents a substantial barrier to establishing a model of flight delays due to the unequal distribution (approximately half of delays are concentrated into two categories). Additionally, larger airports are likely to provide much more flight delay data than smaller ones would, thereby causing bias and skewness in the results produced by the flight delay model being developed. In order to alleviate the effects of such bias and skewness in regard to the model developed from the data provided by airports of varying size, we carry out an exploratory data analysis, give consideration to normalizing the features associated with the delay data, and evaluate the model's performance by way of an assessment of performance over various routes and types of airports. Modelling the complex, nonlinear interactions between factors that produce flight delays is another challenge; factors creating delay include, for example, weather conditions and operational constraints. We have employed both baseline and ensemble type models with the hope of more thoroughly capturing the interaction among the various factors affecting delay, and we have compared the performance of the two types of models based upon multiple evaluation metrics in order to promote stability in our model results.

IV. Expected Outcomes

Based on previous research findings and the characteristics of the BTS data set, ensemble models are expected to provide better predictions of the probability and duration of delays than baseline models. In addition, based on the analysis performed to date, it is anticipated that there will be only a few key factors contributing to overall delay patterns, these factors are likely to include late aircraft delays and carrier delays while climate-based delays may fluctuate more by geographic location and season. Moreover, the results of prediction analyses and delay factor analyses should provide valuable airline selection recommendations to users through the application's ability to identify the least risky travel options. Overall, this project seeks to demonstrate that integrating predictive modelling with interpretable delay factors improves the accuracy of the predictions, as well as the usefulness of the resulting decision support.

V. Team Member Contribution

Each team member contributes to both the technical and writing components of the project, with clearly defined responsibilities across front-end and back-end development.

A. Technical Components

Yufan Peng is responsible for data collection, model development, and model evaluation. Zhiying Huo focuses on exploratory data analysis (EDA) and back-end implementation. Xinrui Song contributes to model development and evaluation. Lin Wu is responsible for front-end development and supports exploratory analysis. Xinyi Huang focuses on literature review, exploratory data analysis, and model evaluation. This distribution ensures a balanced workflow across data processing, modeling, evaluation, and application development.

B. Writing Components

Yufan Peng is responsible for drafting the Introduction and Machine Learning Pipeline sections. Zhiying Huo contributes to the Data and Preprocessing description within the pipeline section. Xinrui Song focuses on the Model Training and Evaluation sections. Lin Wu is responsible for the Front-End (Streamlit application) description. Xinyi Huang contributes to the Background section and supports the Risk & Mitigation and Expected Outcomes sections. All team members collaboratively review and revise the final report to ensure clarity, consistency, and completeness.